

NEURAL NETWORK (PERCEPTRON)

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In

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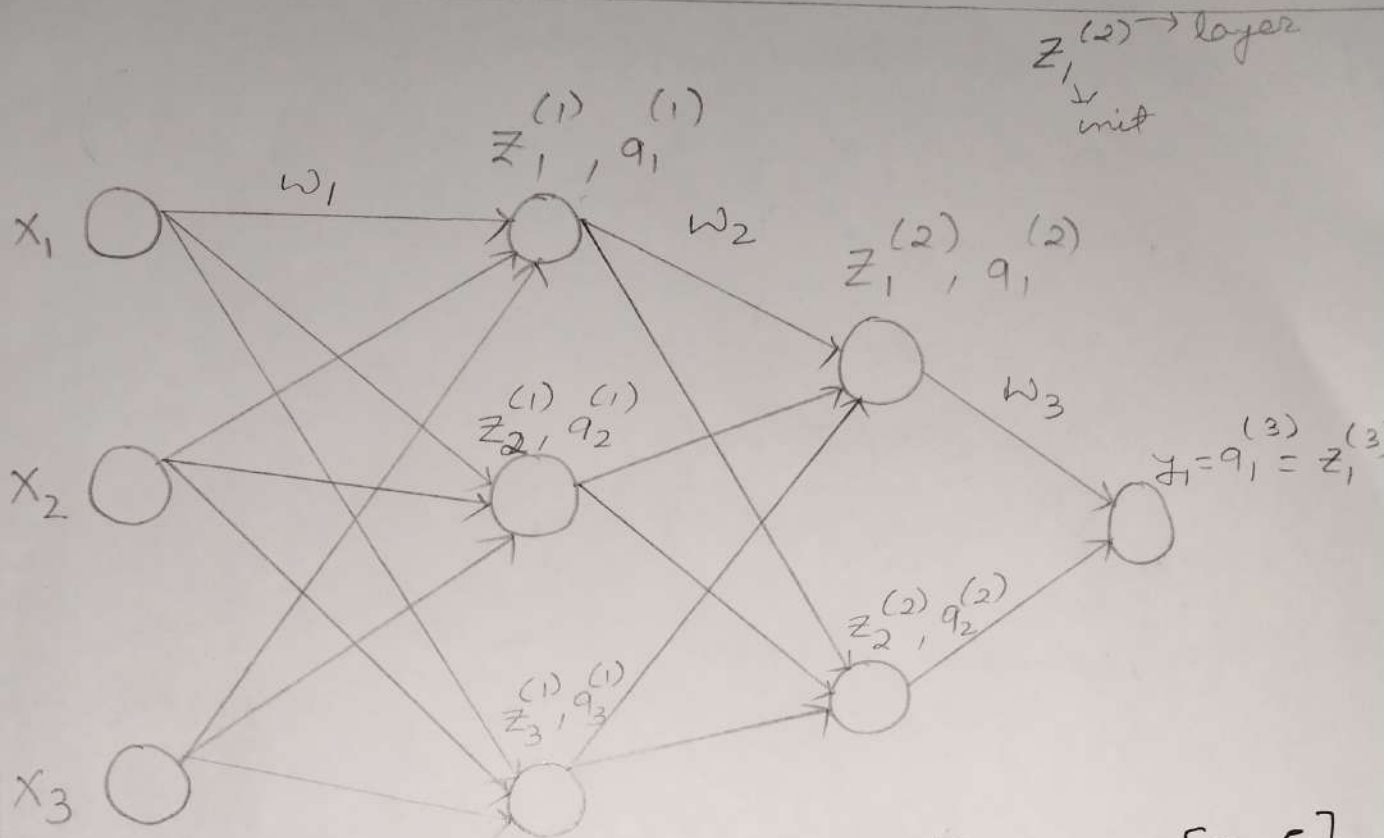
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①

Q.2



$$w_1 = \begin{bmatrix} 1 & 1 & 2 \\ 3 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix}, w_2 = \begin{bmatrix} 1 & 1 & 2 \\ 3 & 1 & 1 \end{bmatrix}, w_3 = \begin{bmatrix} 2 & 5 \end{bmatrix},$$

$$x = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \text{ Loss} = (\hat{y} - y)^2, \text{ Actual output} = y = 10$$

For the above network, find:-

- ① $\frac{\partial L}{\partial w_3}$ and sum of all elements in $\frac{\partial L}{\partial w_3}$
- ② $\frac{\partial L}{\partial w_2}$ and sum of all elements in $\frac{\partial L}{\partial w_2}$
- ③ $\frac{\partial L}{\partial w_1}$ and sum of all elements in $\frac{\partial L}{\partial w_1}$

(2)

Ans) Forward Propagation $\longrightarrow$

$$z^{(1)} = w_1 x$$

$$a^{(1)} = \sigma(z^{(1)})$$

$$z^{(2)} = w_2 a^{(1)}$$

$$a^{(2)} = \sigma(z^{(2)})$$

$$z^{(3)} = w_3(a^{(2)}) = \hat{y}$$

$\hookrightarrow$ this is because the last layer, i.e., output layer has linear activation.

Solving all the terms, which we are going to use in the forward pass:-

$$z^{(1)} = w_1 x$$

$$= \begin{bmatrix} 1 & 1 & 2 \\ 3 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}$$

$3 \times 3 \quad \quad 3 \times 1 \quad \quad 3 \times 1$

$$a^{(1)} = \sigma(z^{(1)})$$

$$= \sigma \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix} = \begin{bmatrix} 0.982 \\ 0.993 \\ 0.997 \end{bmatrix}$$

(3)

$$z^{(2)} = w_2 a^{(1)}$$

$$= \begin{bmatrix} 1 & 1 & 2 \\ 3 & 1 & 1 \end{bmatrix}_{2 \times 3} \begin{bmatrix} 0.982 \\ 0.993 \\ 0.997 \end{bmatrix}_{3 \times 1} = \begin{bmatrix} 3.969 \\ 4.936 \end{bmatrix}_{2 \times 1}$$

$$a^{(2)} = \sigma(z^{(2)})$$

$$= \sigma \begin{bmatrix} 3.969 \\ 4.936 \end{bmatrix} = \begin{bmatrix} 0.981 \\ 0.992 \end{bmatrix}_{2 \times 1}$$

$$w_3 a^{(2)} = z^{(3)} = a^{(3)} = \hat{y} -$$

$$\begin{bmatrix} 2 & 5 \end{bmatrix}_{1 \times 2} \begin{bmatrix} 0.981 \\ 0.992 \end{bmatrix}_{2 \times 1} = \begin{bmatrix} 6.922 \end{bmatrix}_{1 \times 1}$$

Therefore,

$$\begin{aligned} L &= (\hat{y} - y)^2 \\ &= (6.922 - 10)^2 \\ &= 9.45 \quad \text{--- (i)} \end{aligned}$$

Solving Question (i) $\frac{\delta L}{\delta w_3}$

$$\frac{\delta L}{\delta w_3} = \frac{\delta L}{\delta \hat{y}} \times \frac{\delta \hat{y}}{\delta w_3} \quad [\text{chain rule method}]$$

$$\begin{aligned} &= 2(\hat{y} - y) a^{(2)} \\ &= 2(\hat{y} - y)^2 a^{(2)} \\ &= 2(\hat{y} - y) a^{(2)} \end{aligned}$$

(4)

$$2(6.922 - 10) \begin{bmatrix} 0.981 \\ 0.992 \end{bmatrix} = \begin{bmatrix} -6.039 \\ -6.106 \end{bmatrix}$$

Now, sum of all elements in $\frac{\delta L}{\delta w_3}$

$$= (-6.039) + (-6.106)$$

$$\frac{\delta L}{\delta w_3} = -12.145$$

(ii) $\frac{\delta L}{\delta w_2}$

$$= \frac{\delta L}{\delta \hat{y}} \cdot \frac{\partial \hat{y}}{\partial a^{(2)}} \cdot \frac{\partial a^{(2)}}{\partial z^{(2)}} \cdot \frac{\partial z^{(2)}}{\partial w_2}$$

$$\underbrace{2(\hat{y} - y)}_{1 \times 1} \underbrace{w_3}_{1 \times 2} \underbrace{\sigma(z_2)(1 - \sigma(z_2))}_{2 \times 1} \underbrace{a^{(1)}}_{3 \times 1}$$

we have to arrange in the order of matrix.

$$\underbrace{w^{(3)T}}_{2 \times 1} \odot \underbrace{\sigma(z^{(2)})(1 - \sigma(z^{(2)}))}_{2 \times 1} \cdot \underbrace{2(\hat{y} - y)}_{1 \times 1} \cdot \underbrace{a^{(1)T}}_{1 \times 3}$$

$$\underbrace{\hspace{10em}}_{2 \times 1}$$

$\Rightarrow$ 2×3 matrix order.

(5)

$$\begin{bmatrix} -0.227 \\ -0.215 \end{bmatrix} \begin{bmatrix} 0.982 & 0.993 & 0.997 \end{bmatrix}$$

$$= \begin{bmatrix} -0.223 & -0.225 & -0.226 \\ -0.211 & -0.213 & -0.214 \end{bmatrix} = \frac{\delta L}{\delta \omega_2}$$

2×3

Now, sum of all the elements for $\frac{\delta L}{\delta \omega_2}$

$$= \{ (-0.223) + (-0.225) + (-0.226) + (-0.211) + (-0.213) + (-0.214) \}$$

$$= \underline{\underline{-1.312}}$$

Free $\frac{\delta L}{\delta \omega_1}$

$$\rightarrow \frac{\delta L}{\delta \hat{y}} \times \frac{\delta \hat{y}}{\delta a^{(2)}} \times \frac{\delta a^{(2)}}{\delta z^{(2)}} \times \frac{\delta z^{(2)}}{\delta a^{(1)}} \times \frac{\delta a^{(1)}}{\delta z^{(1)}} \times \frac{\delta z^{(1)}}{\delta \omega_1}$$

$$= 2(\hat{y} - y) \omega_3 \sigma z^{(2)} (1 - \sigma z^{(2)}) \omega_2 \sigma z^{(1)} (1 - \sigma z^{(1)}) \omega_1$$

we have to arrange the eqn in the order of ω_1 matrix.

$$\{ \omega_1 \odot \sigma(z^{(1)} (1 - \sigma z^{(1)})) \} \left[2(\hat{y} - y) \sigma(z^{(2)} (1 - \sigma z^{(2)}))^T \odot \omega_3 \right] \omega_2$$

(6)

$$= \left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \odot \begin{bmatrix} 0.017 \\ 0.006 \\ 0.002 \end{bmatrix} \right\} \cdot \left\{ \begin{bmatrix} -0.110 & -0.043 \end{bmatrix} \odot \begin{bmatrix} 1 & 1 & 2 \\ 3 & 1 & 1 \end{bmatrix} \right\}$$

$$= \begin{bmatrix} 0.017 \\ 0.006 \\ 0.002 \end{bmatrix}_{3 \times 1} \cdot \left\{ \begin{bmatrix} -0.022 & -0.215 \end{bmatrix}_{1 \times 2} \begin{bmatrix} 1 & 1 & 2 \\ 3 & 1 & 1 \end{bmatrix}_{2 \times 3} \right\}$$

$$= \begin{bmatrix} 0.017 \\ 0.006 \\ 0.002 \end{bmatrix}_{3 \times 1} \begin{bmatrix} -0.667 & 0.237 & 0.259 \end{bmatrix}_{1 \times 3}$$

$$= \begin{bmatrix} -0.0113 & 0.0040 & 0.0044 \\ -0.0040 & 0.0014 & 0.0015 \\ -0.0013 & 0.0004 & 0.0005 \end{bmatrix}_{3 \times 3}$$

$$\rightarrow \frac{\delta L}{\delta \omega_1}$$

Sum of all the elements for $\frac{\delta L}{\delta \omega_1}$

$$= \{ (-0.0113) + (0.0040) + (0.0044) + (-0.0040) + (0.0014) + (0.0015) + (-0.0013) + (0.0004) + (0.0005) \}$$

$$= -0.1061$$